Uwayo Pascaline

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EDUCATION

Columbia University - The Fu Foundation School of Engineering and Applied Science - New York, NY | *B.S. Computer Science, Expected 2030*

- Selected coursework: Calculus I, Introductory Mechanics & Thermodynamics, Principles of Economics, University Writing, The Art of Engineering.

Rwanda Coding Academy - Rwanda | *Advanced Level Software Programming & Embedded Systems | 2023–2026*

- Intensive training in software development, programming, embedded systems, databases, and technology-driven problem solving.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C/C++, PHP, Kotlin, Swift, SQL, R, HTML/CSS, perl
Web Development: React, Next.js, Vue.js, Node.js, Express.js, FastAPI, Spring Boot, Laravel, REST APIs, WebSockets, Tailwind CSS, Bootstrap
Mobile Development: React Native, Flutter, Kotlin, Swift, SwiftUI, Android, iOS, Expo
AI/ML & Data Science: TensorFlow, PyTorch, Keras, OpenCV, ONNX Runtime, ArcFace, MediaPipe, Pandas, NumPy, Data Analysis, Statistical Analysis, Data Visualization, Data Pipelines, OpenAI API, Matlab
Data & BI: Tableau, Data Visualization, Data Analytics, Statistical Computing, SQL, Data Cleaning, Exploratory Data Analysis
Databases: PostgreSQL, MySQL, MariaDB, MongoDB, Database Design, Data Modeling, JSON/JSONB
DevOps & Tools: Docker, Kubernetes, Git, GitHub, Jenkins, CI/CD, Linux/Kali Linux, Figma, VS Code
Cybersecurity: Nmap, Wireshark, Burp Suite, Metasploit, Masscan, Recon-ng, PicoCTF, Penetration Testing
Embedded & IoT: Arduino, ESP32, LoRa/RFM95, Sensors, Robotics, Embedded Systems, Computer Vision
Research skills: Research Methodology, Literature Review, Data Collection, Quantitative Data Analysis, Statistical Analysis, Computational Research, Critical Thinking, Scientific Writing, Data Interpretation, Research Presentation

PROFESSIONAL EXPERIENCE

Full-Stack Developer Intern - SACOLA | *Summer 2025*

- Faced feature and integration issues across **4 application modules**; developed and debugged frontend, backend, API, and database components, reducing development bottlenecks and improving application functionality.
- Supported **50+ endpoints** through testing and debugging, improving reliability and enabling smoother user workflows.

Backend Developer - New Health Intelligence Center | *Sep 2025–Aug 2026*

- Supported healthcare systems handling **100000+ records/requests**; developed backend services, APIs, and database components to improve data processing and accessibility.
- Addressed backend performance and integration challenges across **15+ services**, improving reliability of digital health-data workflows.

DevOps Engineer - RCA Management Information System (RCA MIS) | *Nov 2024–May 2026*

- Supported a school platform serving **500+ users** that faced server bottlenecks during concurrent access; optimized deployments, server resources, and system performance to improve reliability during peak usage.
- Managed infrastructure for **5 services** handling student records, courses, grades, staff records and financial data; strengthened security, scalability, and system availability.

LEADERSHIP & ACTIVITIES

Minister of Sports, Culture & Entertainment - Rwanda Coding Academy Student Council | *2025–2026*

- Coordinated **12+ student activities** for a school community of **~500 students**, improving participation and communication across sports and cultural events.

Vice Captain & Team Organizer - Rwanda Coding Academy Basketball Team | *High School*

- With no official coach, organized practices, communication, and game-day responsibilities for **21 players**, keeping the team coordinated and active.

Soccer & Basketball Player - School Athletics | *High School*

- Competed in **2 sports** while balancing academics, strengthening teamwork, discipline, and time management.

Peer Mentor - Brainiacs, Rwanda Coding Academy | *2024–2026*

- Mentored **100+ new students** in programming and study strategies, helping them adapt to the academy's technical environment.

Student Researcher - Rwanda Coding Academy | *2024–2026*

- Conducted **3 research projects** under Dr. Awet Fesseha, applying AI, engineering, and data science to real-world problems and producing **3 research papers**.

SELECTED PROJECTS

SurVie - Health Intelligence Platform - Team Project | *demo* | *2025–2026*

- Designed a health platform concept connecting hospital data into an ecosystem that can provide users with a more unified view of their medical information.
- Explored personalized AI capabilities and biometric-based emergency identification to support faster access to relevant health information.

Stratiq - AI & Data Science for Financial Decision-Making - Personal | *demo* | *2025–2026*

- Developed a project focused on using data science and AI to support financial prediction, analysis, and decision-making for companies.

OpenLab Virtual STEM Laboratory - Team Project | *2025–2026*

- Developing a virtual laboratory platform designed to expand access to STEM experimentation for rural communities and schools with limited laboratory resources.
- Combines software and interactive learning concepts to make practical STEM experiences more accessible.

CERTIFICATIONS

- **App Development with Swift** - Apple
- **Ideation and Prototyping** - CODE University of Applied Sciences
- **RCA Hackathon, 4th Edition** - KOICA
- **Introduction to Cybersecurity** - ThinkCyber LTD
- **Penetration Testing** - ThinkCyber LTD
- **Network Research** - ThinkCyber LTD
- **Linux Fundamentals** -ThinkCyber LTD
- **Python Fundamentals** - ThinkCyber LTD

ADDITIONAL INFORMATION

Languages: English, Kinyarwanda, French

Non technical skills: Leadership & Team Coordination, Communication, Teamwork & Collaboration, Problem Solving, Project Management, Mentorship, Organization & Time Management, Adaptability, Initiative, Presentation